Quantum Mechanics Revision and Course Overview

> *This session focused on revising **Quantum Mechanics** (Unit 1) in Engineering Physics, covering essential topics and their importance for university exams.*

Course and Semester Information

Semester	Subjects for Computer Science Students	Subjects for All Students
3rd	Discrete Mathematics (DSTL)	Engineering Mathematics III & IV

- Gateway Classes app will provide 2nd-year courses after the last 1st-year exam.
- Recommended to share materials with juniors or relatives for better preparation.

Syllabus Topics for Quantum Mechanics Unit

Topic	Importance	Exam Notes
Limitations of Classical Mechanics	Important (2 marks)	Basics and differences with Quantum Mechanics
Planck's Theory of Black Body Radiation (Qualitative)	Important	Theory focus; derivations and numericals given but only theory explained here
Compton Effect	Very Important	Derivation, shift calculation, numericals
De Broglie Concept of Matter Wave	Average Importance	Numericals often asked
Davisson-Germer Experiment	Important	2 to 7 marks question
Phase Velocity and Group Velocity	Important	Derivation and relation, 2 or 7 marks
Time-dependent and Independent Schrödinger Wave Equation	Very Important	Frequently repeated in exams
Physical Interpretation of Wave Function	Important (2 to 7 marks)	Explanation of wave function's physical meaning
Particle in One-Dimensional Box	Very Important	Frequently repeated, detailed study

Difference Between Classical and Quantum Mechanics

Aspect	Classical Mechanics	Quantum Mechanics
Objects studied	Macroscopic (large size) objects	Microscopic (very small) objects
Based on	Newton's laws of motion	Schrödinger wave equation
Predictability of future state	If position known, future is predictable	Future can't be precisely predicted; uncertainty principle applies
Nature of light	Wave nature only	Dual nature: wave and particle (photon)
Explains phenomena	Cannot explain phenomena like black body radiation, Compton effect, photoelectric effect, Raman effect, atomic stability	Explains all above phenomena

Limitations of Classical Mechanics

- Cannot explain behavior of microscopic objects.
- Fails to explain many observed phenomena:
  - Black body radiation spectrum
  - Compton effect
  - Photoelectric effect
  - Raman effect- Cannot explain atomic stability.
- Cannot explain variation in specific heat of metals and gases.
- These limitations led to the development of Quantum Mechanics.

Black Body and Black Body Radiation

> ***Perfect black body** absorbs all incident radiation of all wavelengths and emits radiation when heated.*

- Real bodies are not perfect black bodies; perfect black bodies do not exist in nature.
- Black bodies are good absorbers and emitters of radiation.
- Types of black bodies:
  - **Natural:** Lamp black, platinum black, sun, stars.
  - **Artificial:** Electric heater, ferrous black body.- Color is **not necessarily black**; color depends on visible light absorption only.
- **Black body spectrum:**
  - Intensity of radiation increases with wavelength, reaches a maximum (λ_{max}) then decreases.
  - λ_{max} shifts toward shorter wavelengths with increasing temperature.
  - Energy distribution is non-uniform.

- Total emitted energy increases with temperature.

Important Laws Related to Black Body Radiation

Law	Statement	Applicability & Limitations
Stefan's Law	Total radiant energy per unit area per unit time $\propto T^4$ (temperature to the fourth power)	No numericals asked; classical mechanics domain
Wien's Displacement Law	Wavelength of maximum intensity λ_m inversely proportional to absolute temperature:	

$$\lambda_m \propto \frac{1}{T}$$

Important for 2 marks questions		
Wien's Radiation Formula	Amount of radiant energy emitted per unit area, per unit time, per wavelength range at given temperature (formula given in syllabus)	Applicable only for shorter wavelengths; classical limitation
Rayleigh-Jeans Law	Used classical electromagnetism and thermodynamics to calculate spectral energy of black body radiation	Applicable only for longer wavelengths; classical limitation

Planck's Quantum Hypothesis

- In 1900, Max Planck proposed that radiation consists of tiny packets (quanta) of energy, **photons**.
- Energy of each photon:

$$E = h\nu$$

where h is Planck's constant and ν is frequency.

- Contrary to classical assumption of continuous energy emission, energy is emitted and absorbed in discrete packets.
- Assumptions for deriving Planck radiation formula:

1. Atoms on black body walls behave as oscillators with characteristic frequencies.
2. Oscillators emit/absorb energy only in discrete packets (quanta).
3. Energy of oscillators is quantized:

$$E_n = nh\nu \quad \text{where } n = 1, 2, 3, \ldots$$

- Planck's formula resolves all classical limitations and can derive previous laws (Wien, Stefan, Rayleigh-Jeans) as limiting cases.

Compton Effect

- Discovered by H. Compton in 1921: When high frequency monochromatic radiation (X-rays) scatters from a substance, the scattered radiation includes wavelengths shifted to longer values.
- **Definitions:**

$$\Delta\lambda = \lambda' - \lambda$$

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where λ = initial wavelength, λ' = s
cattered wavelength |
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- **Derivation** based on:
  - Collision between X-ray photon and free electron (elastic collision).
  - Conservation of energy and momentum.
  - Final Compton shift formula:

$$\Delta\lambda = \frac{h}{m_0 c}(1 - \cos\theta)$$

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where m₀ = rest mass of electron, c
= speed of light, θ = scattering ang
le.
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- **Key points:**
  - Wavelength of scattered photon is greater than incident photon.
  - Compton shift depends only on scattering angle, not on material.
  - Explains why Compton effect is not observed in visible spectrum due to very small wavelength change.- **Numericals** involve calculating scattered wavelength and electron kinetic energy using shift formula.

De Broglie Concept of Matter Waves

- De Broglie (1924) proposed every moving particle has an associated wave called **matter wave** or **De Broglie wave**.
- Matter waves have wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where p = momentum, m = mass, v = velocity.

- This concept was experimentally confirmed by **Davisson and Germer (1927)** through electron diffraction experiments.
- Distinction between matter waves and electromagnetic waves:

Feature	Matter Waves	Electromagnetic Waves
Associated with	Moving particles	Accelerated charged particles
Charge	No charge	Charged particles
Electric/magnetic fields	No associated electric or magnetic fields	Accompanied by electric and magnetic fields
Travel through vacuum	No	Yes

- Important formulas and expressions for De Broglie wavelength in terms of kinetic energy, potential difference, and charge are crucial for numericals.

Davisson-Germer Experiment

- Provided first experimental evidence of matter waves predicted by De Broglie.
- Setup:
  - Electrons emitted from tungsten filament (heated by low voltage battery).
  - Accelerated and focused by high voltage battery and fine hole.
  - Electrons scattered by nickel crystal.
  - Scattered electrons detected by movable collector connected to galvanometer (measures current).- Observations:
  - Diffraction pattern (interference) indicated wave nature of electrons.
  - Confirmed that matter waves exist and have measurable wavelength.- Objective: To prove wave nature associated with moving particles.

Phase Velocity and Group Velocity

Concept	Definition	Formula
Phase Velocity (v_p)	Velocity of a single monochromatic wave through a medium	$v_p = \frac{\omega}{k}$
Group Velocity (v_g)	Velocity of a wave packet (group of waves with slightly different wavelengths and velocities)	$v_g = \frac{d\omega}{dk}$

- **Wave packet:** Group of waves with slightly different wavelengths and velocities that interfere constructively over a small region.
- Relation between v_g and v_p in dispersive medium:

$$v_g = v_p - \lambda\frac{dv_p}{d\lambda}$$

- In **non-dispersive media**, group velocity equals phase velocity.

Schrödinger Wave Equation

- Describes wave motion of matter (particles associated with matter waves).
- Analogous to Newton's laws describing motion in classical mechanics.
- Two types:
  - **Time-dependent Schrödinger equation** (includes time variable)
  - **Time-independent Schrödinger equation** (no time variable)- **Wave function (ψ)** represents state of particle at any instant.
  - Physical interpretation: Probability amplitude describing likelihood of particle's position.-

Time-dependent equation derivation (1D):

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi$$

- **Time-independent equation (1D):**

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi$$

- For **free particle** ($V = 0$):

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

- $\hbar = \frac{h}{2\pi}$ is the reduced Planck constant.

Particle in One-Dimensional Box

- Particle constrained to move in 1D box between two infinite potential walls at $x = 0$ and $x = l$.
- Potential energy $V(x)$:

$$V(x) = \begin{cases} 0 \& 0 < x < l \\ \infty \& \text{otherwise} \end{cases}$$

- Boundary conditions:

$$\psi(0) = \psi(l) = 0$$

- Time-independent Schrödinger equation inside the box ($V = 0$):

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \quad \text{where } k^2 = \frac{2mE}{\hbar^2}$$

- General solution:

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

- Applying boundary conditions:
  - At $x = 0$, $\psi(0) = B = 0$.
  - At $x = l$, $\sin(kl) = 0 \Rightarrow kl = n\pi$, where $n = 1, 2, 3, \ldots$- Quantized wave numbers:

$$k = \frac{n\pi}{l}$$

- Energy eigenvalues (discrete energy levels):

$$E_n = \frac{n^2\pi^2\hbar^2}{2ml^2} \quad n = 1, 2, 3, \ldots$$

- Eigenfunctions:

$$\psi_n(x) = A\sin\left(\frac{n\pi x}{l}\right)$$

- Normalization condition:

$$\int_0^l |\psi_n(x)|^2 dx = 1$$

- **Numericals** typically ask for lowest permitted energy levels E_1, E_2 etc., using above formula.

Important Formulas and Constants for Numericals

Quantity	Formula / Value
Energy of photon	$E = h\nu$
Momentum of photon	$p = \frac{h\nu}{c}$
Compton shift	$\Delta\lambda = \frac{h}{m_0 c}(1 - \cos\theta)$
De Broglie wavelength	$\lambda = \frac{h}{mv}$ or $\lambda = \frac{h}{p}$
Planck constant (h)	6.63×10^{-34} Js
Electron rest mass (m_0)	9.1×10^{-31} kg
Speed of light (c)	3×10^8 m/s
Electron volt to joule	$1\ \text{eV} = 1.6 \times 10^{-19}$ J

Key Exam Questions to Prepare

- Limitations of classical mechanics.
- Definitions and concepts of black body, black body radiation, Planck's hypothesis.
- Derivations and formulas of Compton effect and Compton shift.
- De Broglie's hypothesis and wavelength formulas.
- Experimental evidence: Davisson-Germer experiment.
- Phase velocity vs group velocity definitions, differences, and relation.
- Time-dependent and time-independent Schrödinger wave equation derivations.
- Physical interpretation of wave function.
- Particle in one-dimensional box: eigenvalues, eigenfunctions, normalization.
- Important formulas and laws: Stefan's law, Wien's laws, Rayleigh-Jeans law, Planck radiation formula.
- Solving numerical problems involving energy, wavelength, and Compton shift.

Important Concepts with Metaphors and Examples

- **Compton Effect Metaphor:** Like a ball striking another ball and transferring energy, an X-ray photon hits an electron, changing the photon's wavelength (energy) and causing recoil in electron.
- **Wave Packet Explanation:** A wave packet is like a group of waves overlapping, creating regions of high and low amplitude due to constructive and destructive interference.
- **Particle in Box:** Imagine a particle trapped in a narrow corridor with infinitely high walls — it can only have certain allowed energies and positions defined by wavefunctions.

Sample Numerical Problem Types

- Calculate Compton shift and scattered wavelength using Compton formula.
- Calculate energy of recoil electron.
- Compute De Broglie wavelength for electrons or alpha particles under given potential difference.
- Calculate lowest energy levels (E_1, E_2) of particle in one-dimensional box.
- Calculate oscillator energy for given frequency and temperature applying Planck or classical assumptions.

This guide compiles all essential points and derivations from the lecture transcript to ensure mastery of Quantum Mechanics Unit 1.

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